IMPLANTATION &
2ND WEEK OF DEVELOPMENT



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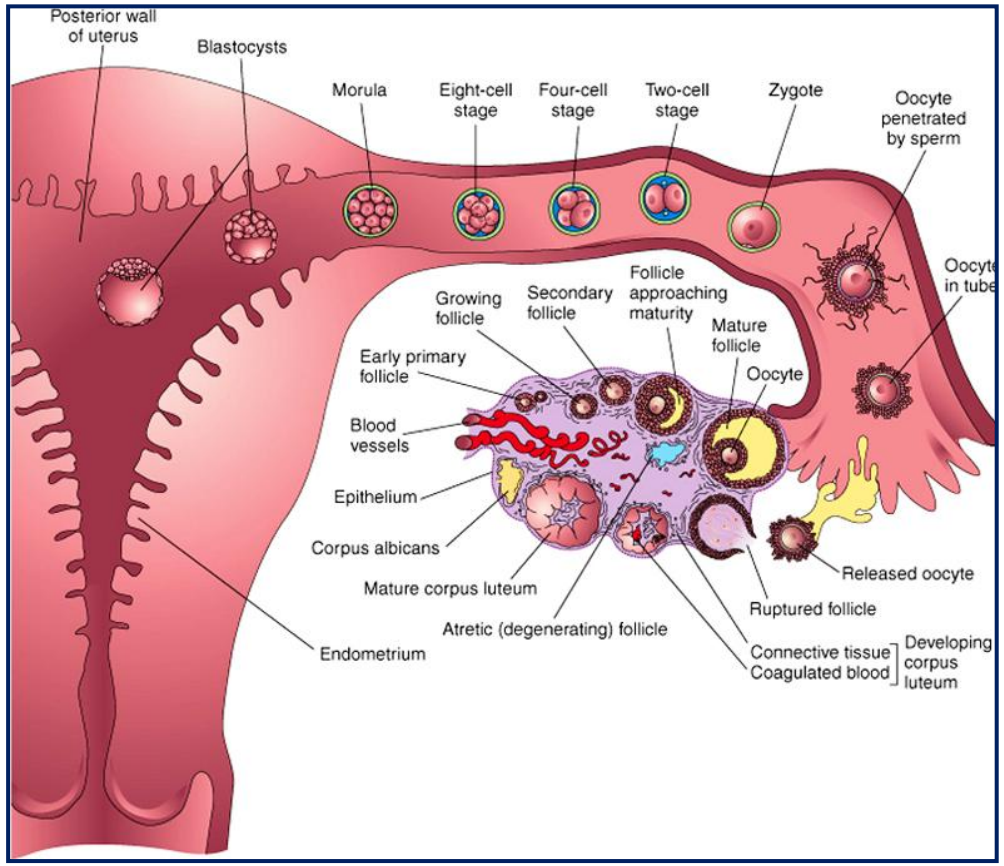
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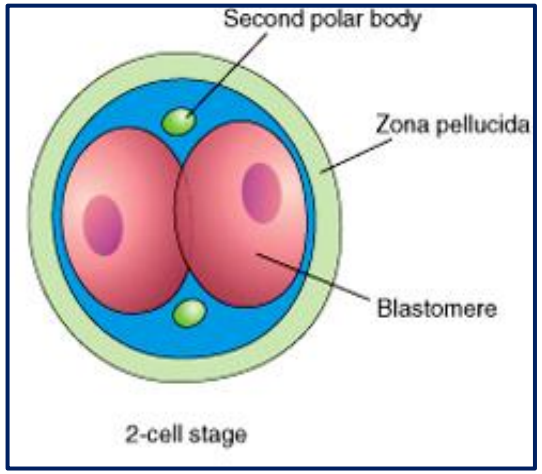
SLOs

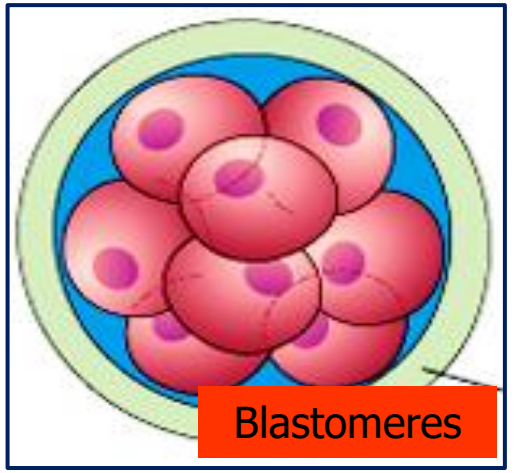
- **SLO# 1:** Describe the normal and abnormal implantation sites
- **SLO# 2:** Describe the major events happening during 2nd week of pregnancy
- **SLO# 3:** Describe the formation of trophoblast
- **SLO# 4:** Describe the development of embryoblast
- **SLO# 5:** Explain the development of extraembryonic mesoderm
- **SLO# 6:** Describe formation of yolk sac, amniotic cavity & uteroplacental circulation

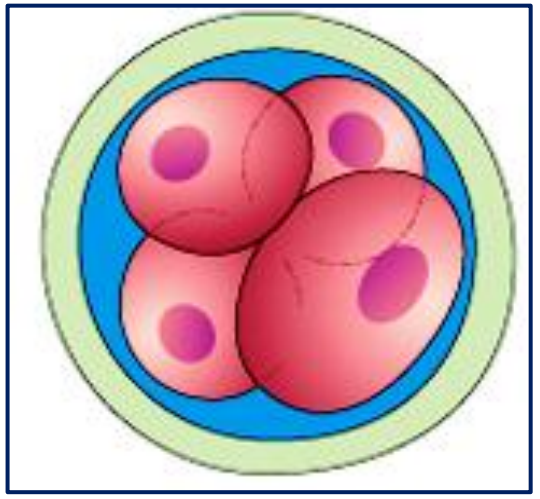
SUMMARY OF FIRST WEEK OF PREGNANCY

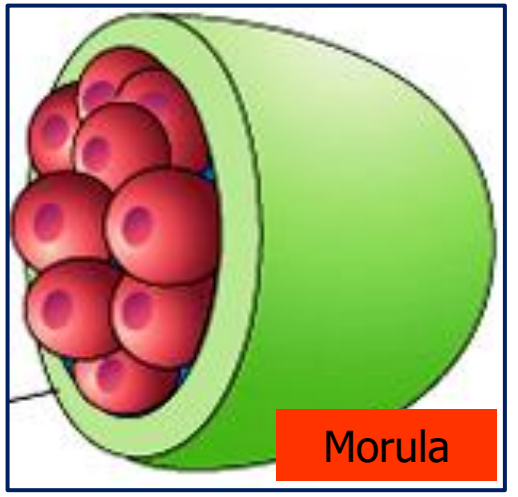
CLEAVAGE





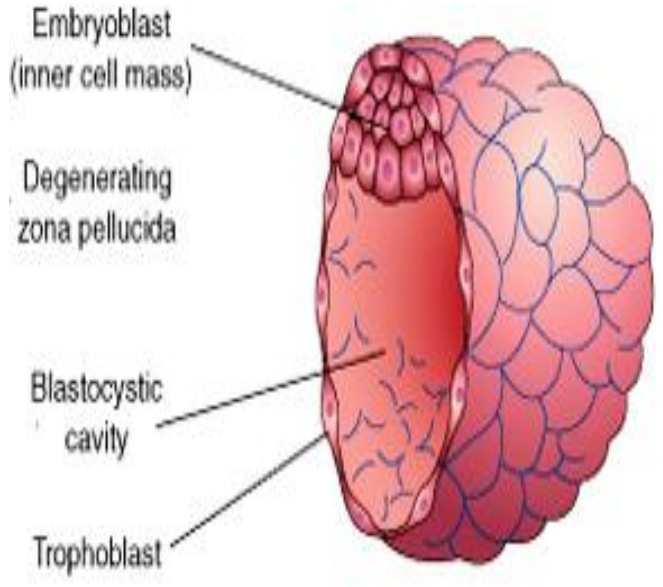






BLASTOCYST

- Shortly after the morula reaches the uterus, fluid passes from the uterine cavity through the zona pellucida into the morula.
- Soon fluid filled spaces appears inside the morula.
- These coalesce & form a cyst – **BLASTOCYST.**
- The fluid filled cavity located inside the blastocyst is called as **BLASTOCOEL.**

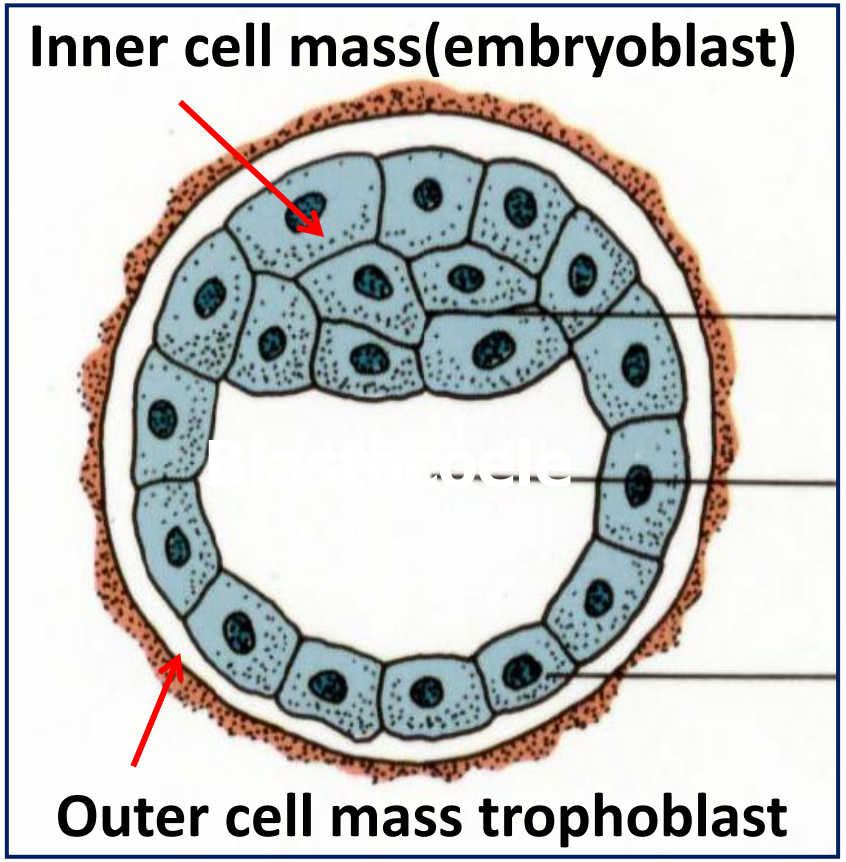


Late blastocyst

Note: The blastocyst is the entire multi-cellular structure in early mammalian development, while the blastocoel is the fluid-filled cavity located within the blastocyst.

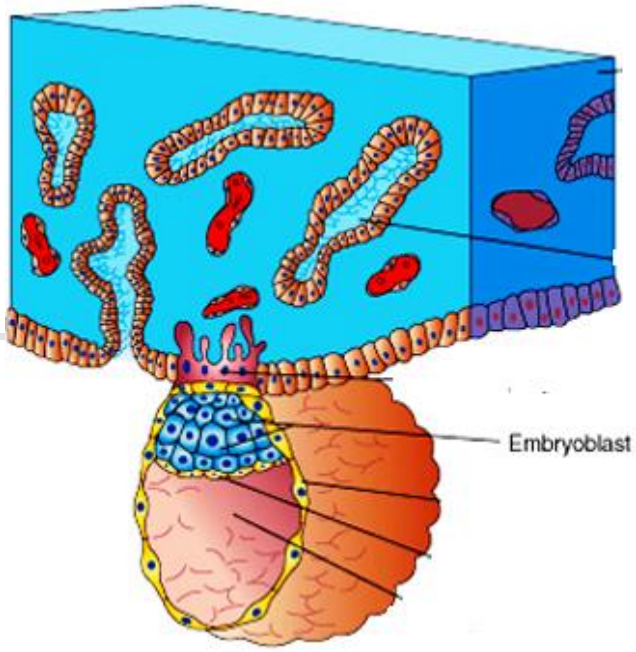
Embryo pole

- As more & more fluid enters the blastocyst it pushes the inner cell mass to an eccentric position.
- The outer layer of flattened cells lining the blastocele – TROPHOBLAST.
- It forms the fetal part of placenta.
- The inner cell mass forms the primordium of the embryo – EMBRYOBLAST



Abembryonic pole

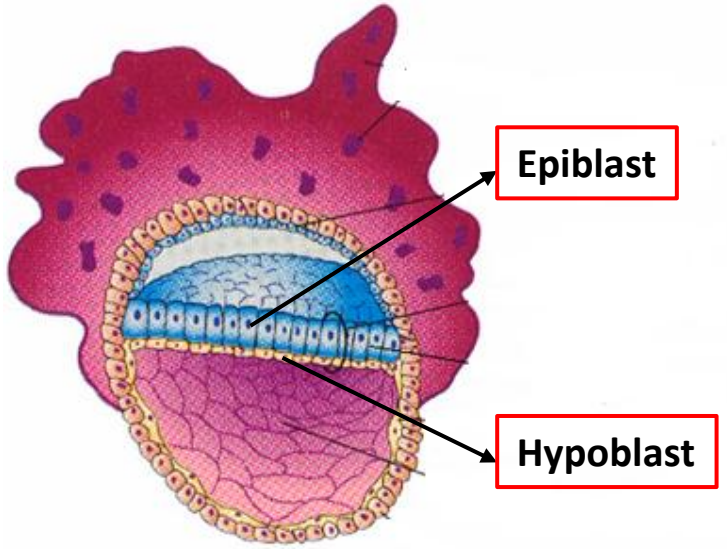
BILAMINAR GERM DISC



- The embryoblast Cells differentiate Into two layers.

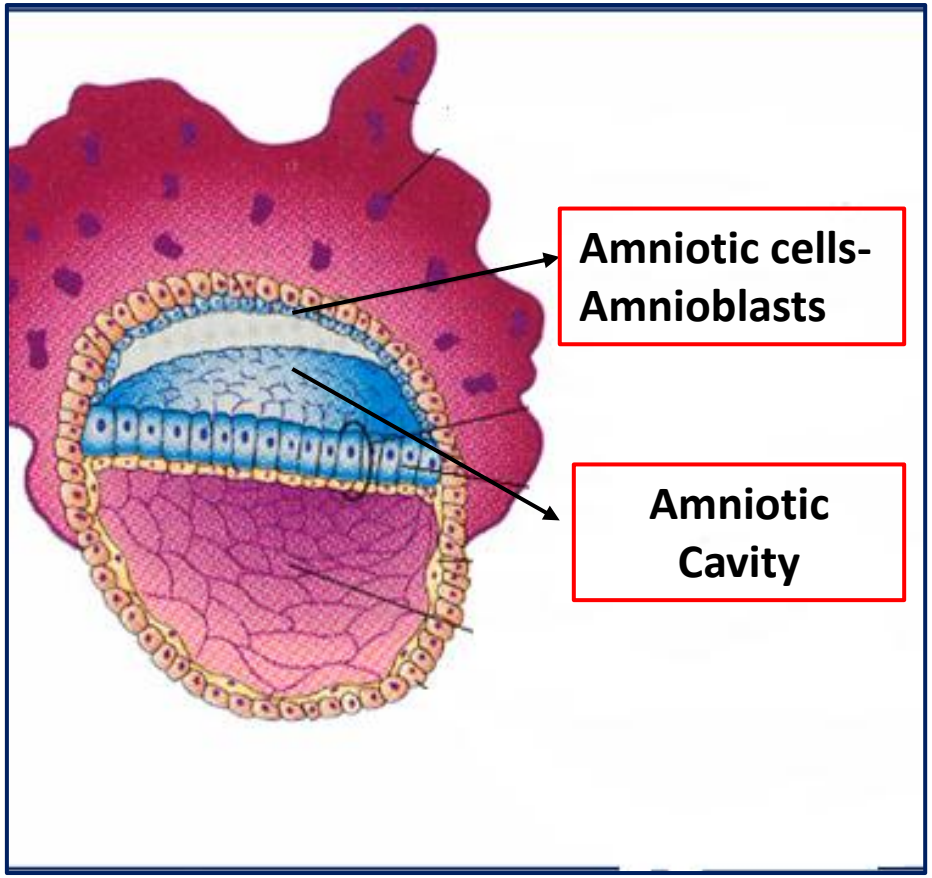
1. **HYPOBLAST :**The cells towards The blastocyst cavity become Cuboidal. (Endoderm).
2. **EPIBLAST:** The cells towards the embryonic pole becomes columnar. (Ectoderm)

Together these form a flat disc **BILAMINAR GERM DISC.**



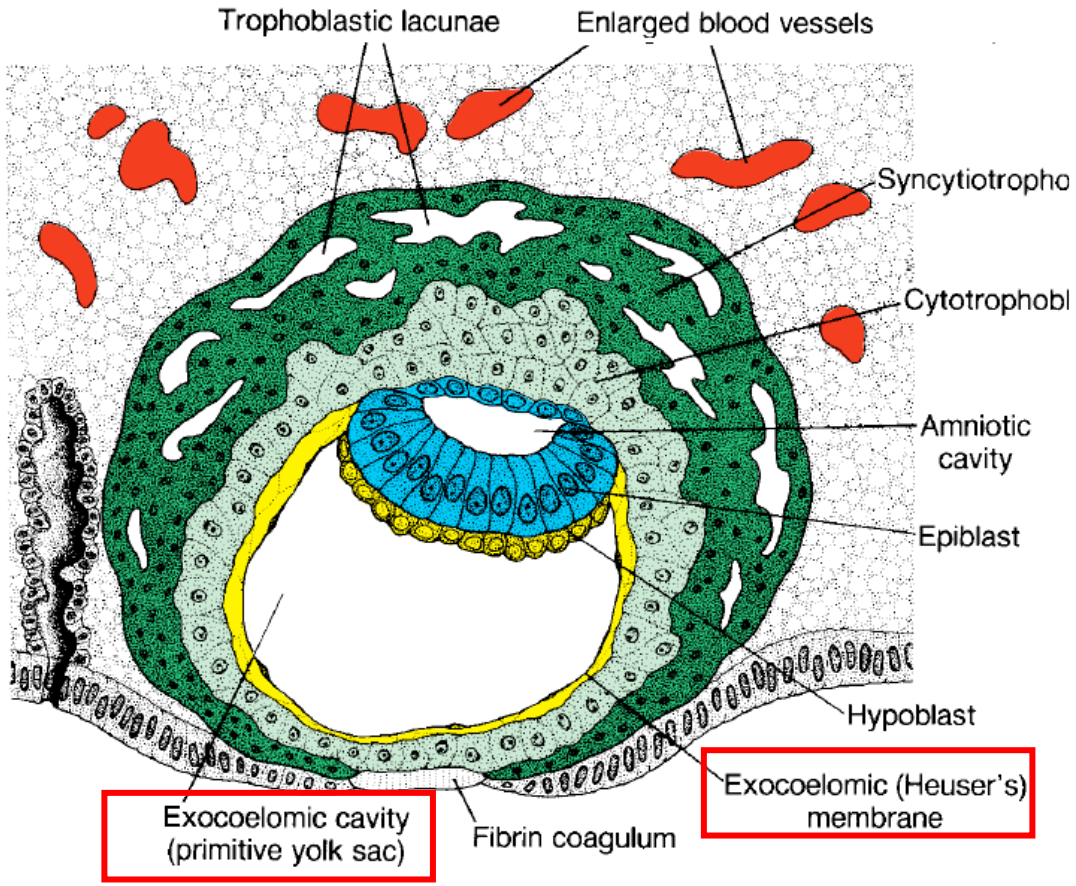
FORMATION OF AMNIOTIC CAVITY

- Soon fluid filled spaces appear between the trophoblast & epiblast – **AMNIOTIC CAVITY**.
- The fluid is called Liquor Amnii Or Amniotic Fluid.
- The amniotic cells get differentiated from the epiblast & line the amniotic cavity - **AMNIOBLASTS**.
- The amniotic cavity is now lined by amnioblasts & its floor is lined by epiblast.



FORMATION OF PRIMARY YOLK SAC

- The hypoblast cells then proliferate & line the inside of the blastocyst cavity – HEUSER'S MEMBRANE.
- The blastocyst cavity is now called as PRIMARY Yolk Sac Or Exo Coelomic Cavity.



CHANGES OCCIRS IN TROPHOBLAST

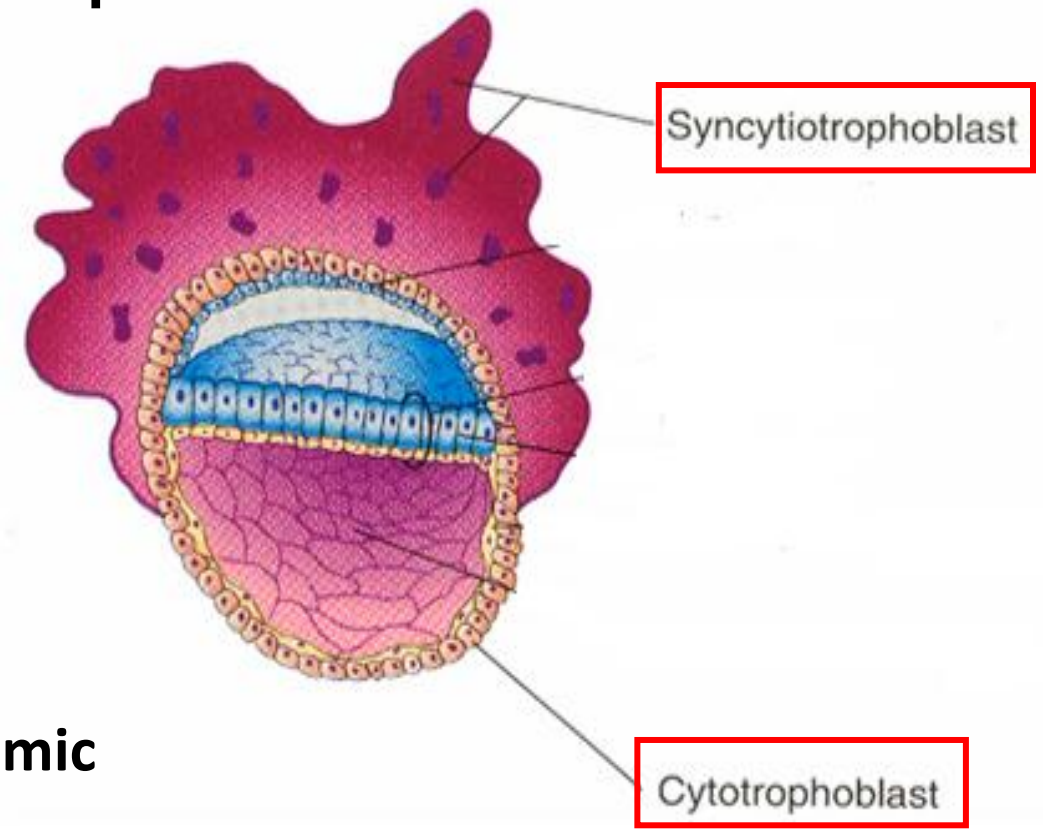
Concurrently the following changes are observed in Trophoblast.

The trophoblast proliferate rapidly.

- **Differentiated into two layers:**

1. **Inner: CYTOTROPHOBLAST**
2. **Outer: SYNCYTIOTROPHOBLAST**

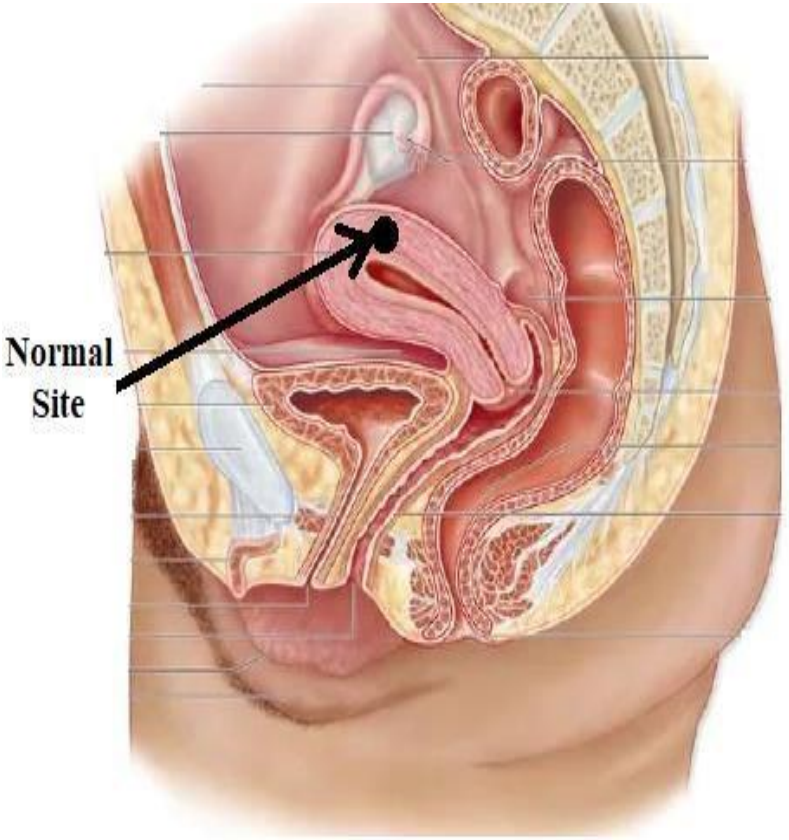
Note: syncytotrophoblast cells loose their cell membrane & form a multinucleated protoplasmic mass.



IMPLANTATION

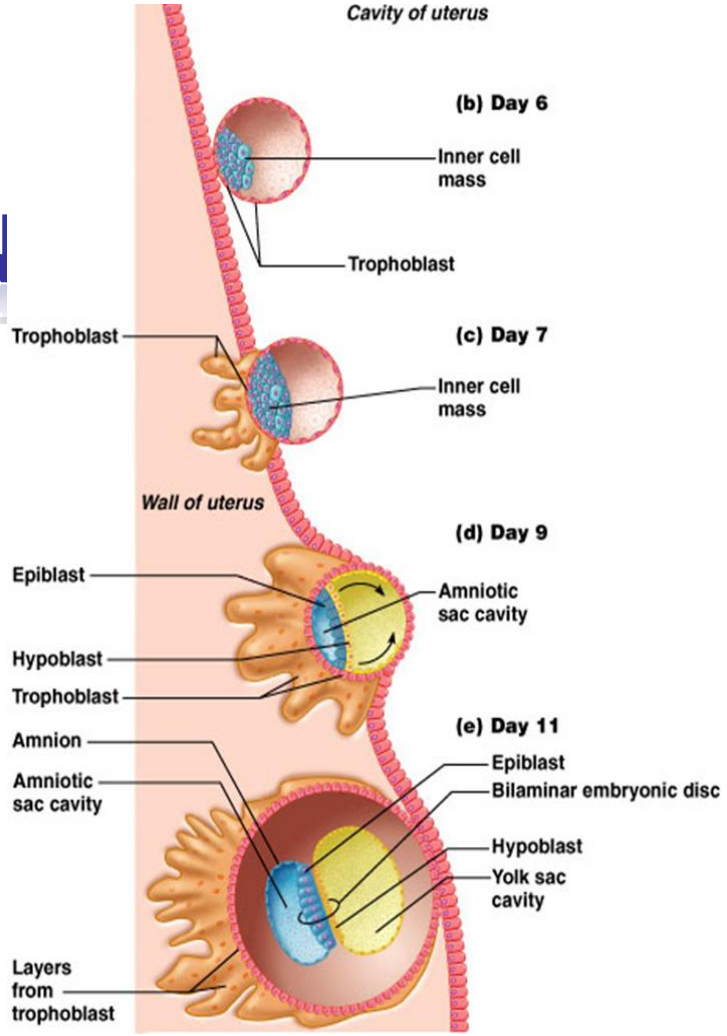
Definition: embedding of **blastocyst** in the functional **endometrium**.

- **TIME**: begins at **end of 1st week (5-6th day)** and is completed by **end of 2nd week (11th – 12th day).**
- **SITE**: **upper** part of **posterior** wall of **body** of uterus (near fundus), near midline



STEPS OF IMPLANTATION

1. Degeneration of zona pellucida (around 5 to 6 days)
2. Exposure of trophoblast (which differentiate into Cyto-trophoblast and inner Outer Syncytio-trophoblast).
3. Numerous finger like processes of the syncytio-trophoblast at the embryonic pole invade the uterine endometrium & start burrowing into it.
4. Implantation begins.



NOTE: The zona pellucida prevents the embryo from sticking to the uterine tube till it reaches the uterine cavity.

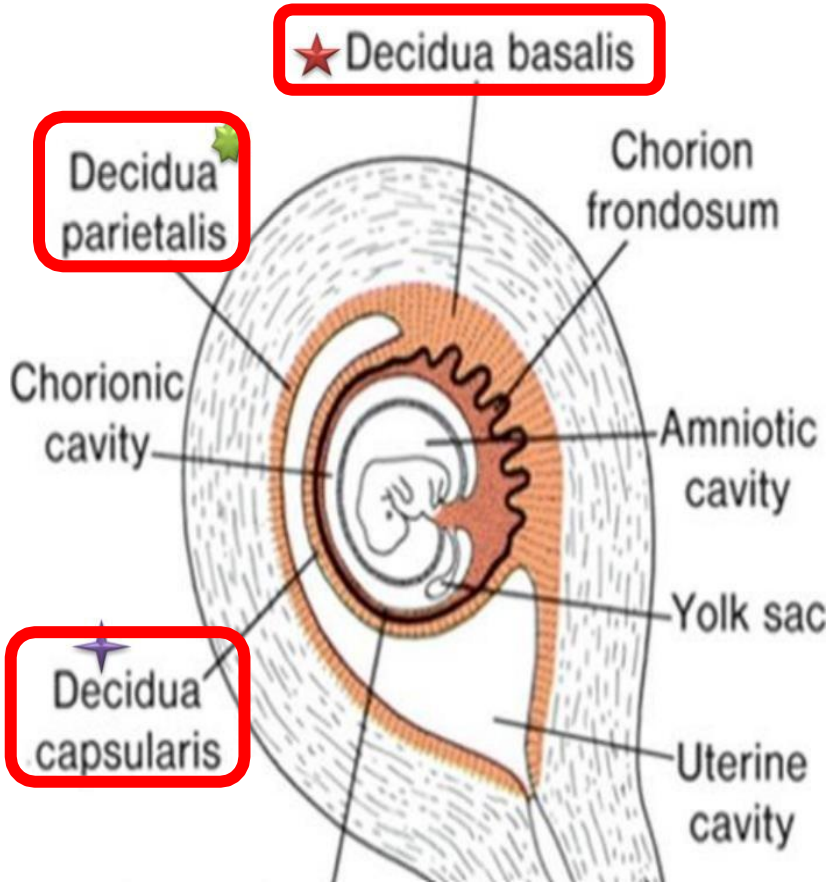
DECIDUA

- **Definition**: endometrium of pregnant uterus.
- **Structure**: secretory phase of endometrium (remember its characters)
- **Control**: Formed under influence of progesterone.
- **Parts**: 3 parts

  1)Decidua basalis: lies between gestation & myometrium.

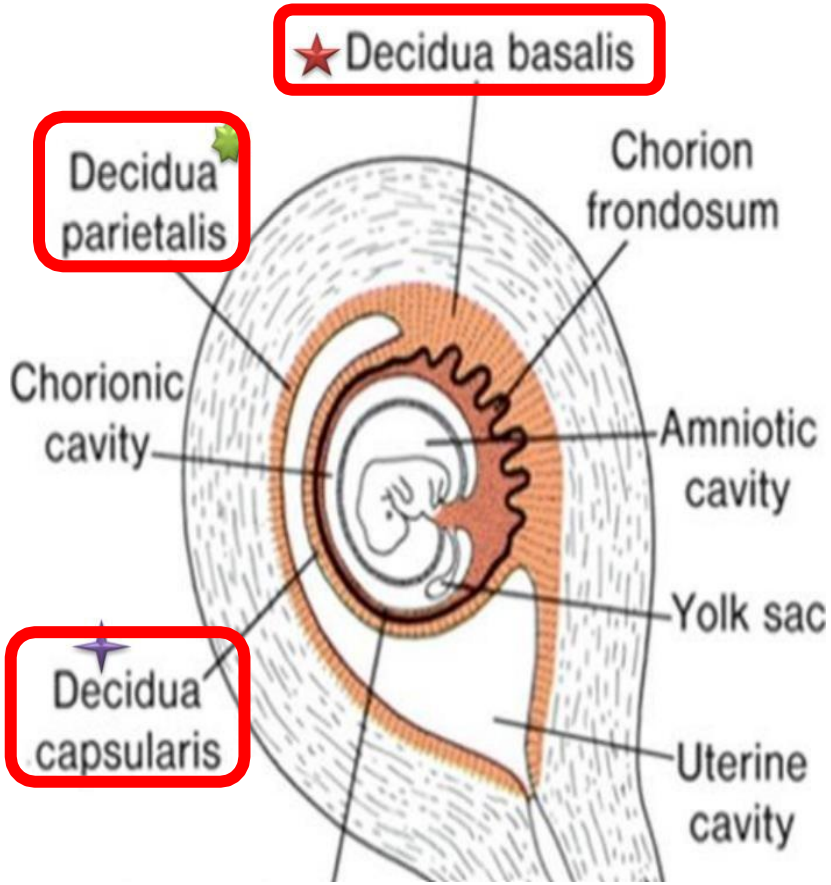
  2)Decidua capsularis: covers gestation.

  3)Decidua parietalis: lines rest of uterine cavity.



DECIDUA CONT.

- **FUNCTION**: Supports the pregnancy.
- **FATE**:
  1. Decidua basalis enlarges to form maternal part of placenta
  2. Decidua capsularis and parietalis → Fuse then degenerate

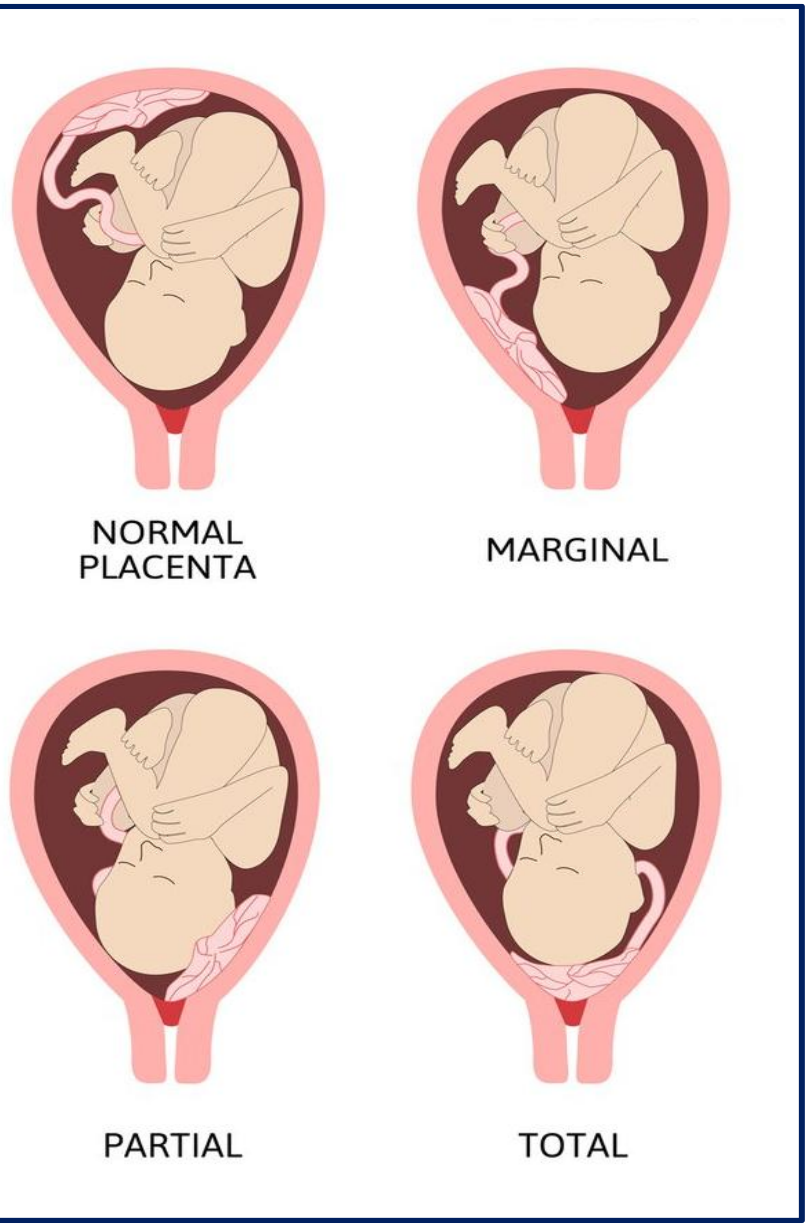


ABNORMAL IMPLANTATION

A. Placenta Previa

- **Definition:** Implantation of **blastocyst** in **inferior** segment of uterus
- **Types:**

1. **Placenta praevia lateralis**: not related to internal os
2. **Placenta praevia marginalis: edge** touches the internal os without covering the opening
3. **Placenta praevia partialis**: Partially covers internal os
4. **Placenta praevia centralis**: completely covers internal os



B) Ectopic Pregnancies (Extra-Uterine):

=Blastocysts implanting outside uterus, may be

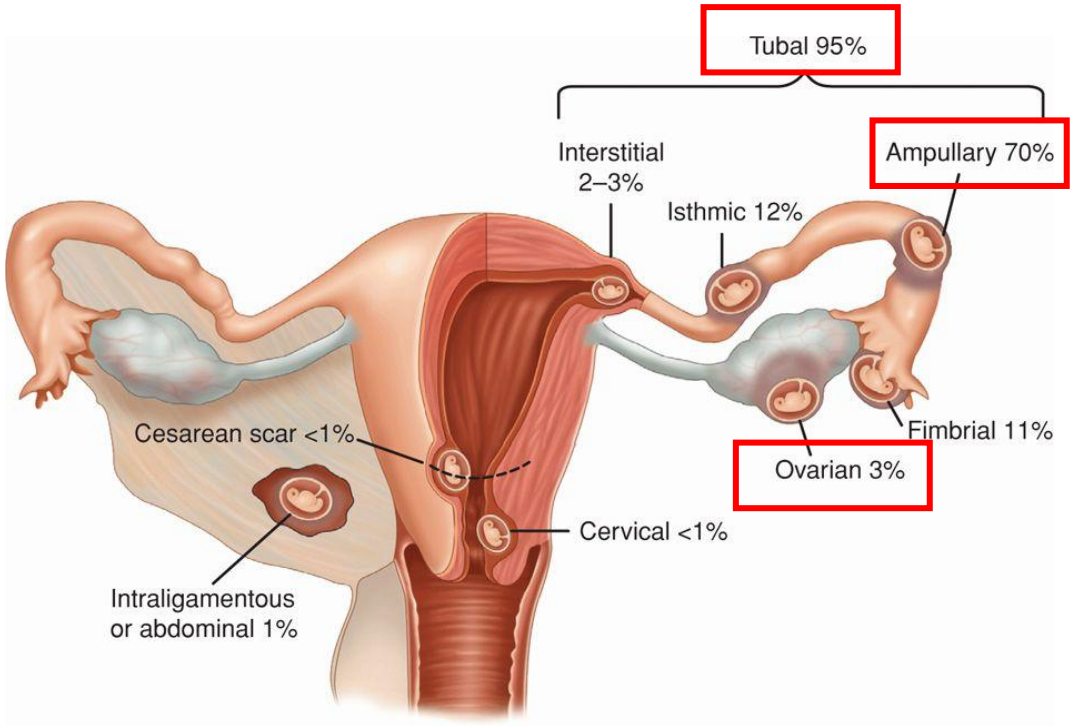
1. Tubal pregnancy (most common-95%):

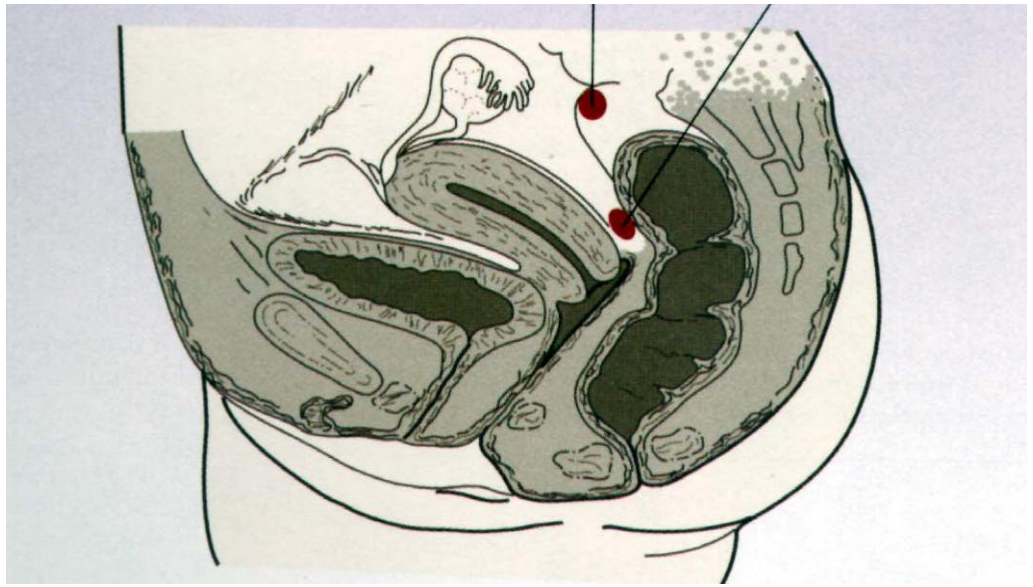
- **Manifestation:** of early pregnancy then pain at side

2. Ovarian pregnancy:

- **Primary:** directly implanted on ovary
- **Secondary**: reimplanted on ovary, after rupture tubal pregnancy.

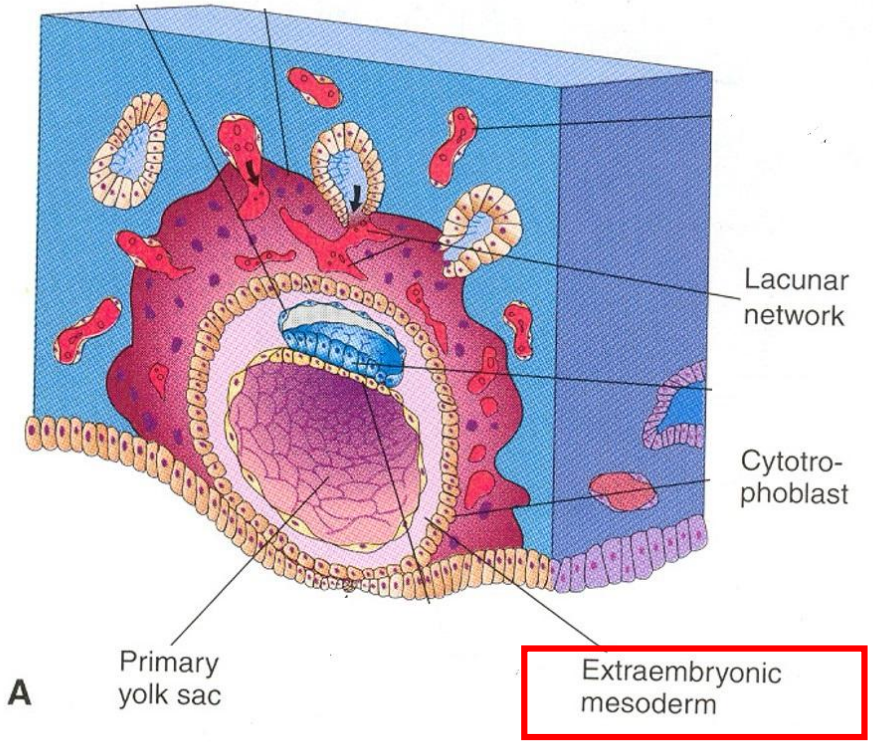
3. Abdominal (Peritoneal) pregnancy: over viscera





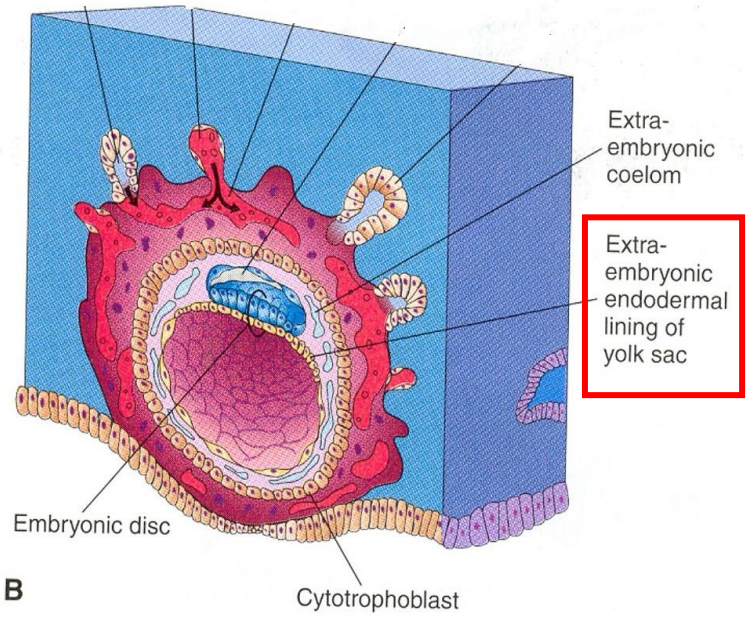
EXTRA EMBRYONIC MESODERM

- A new population of cells are formed between the cytotrophoblast & primary yolk sac – EXTRAEMBRYONIC MESODERM.
- These cells proliferate rapidly & separate the amniotic cavity & yolk sac from trophoblast.

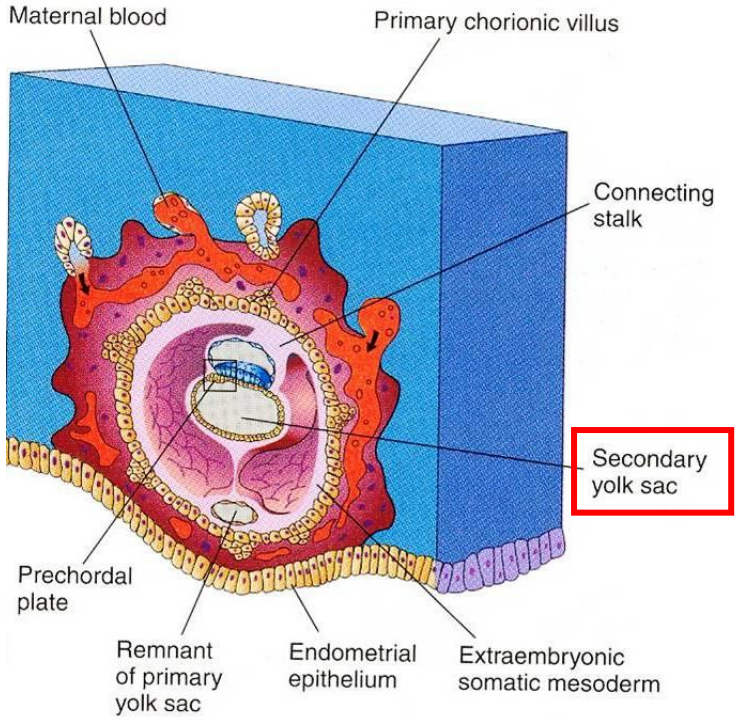


EXTRA EMBRYONIC COELOMIC CAVITY

- Large fluid filled spaces now appear within the extra embryonic mesoderm.
- These coalesce to form **EXTRA EMBRYONIC COELOMIC CAVITY or CHORIONIC CAVITY.**
- Divided into two layers.
  A. **Somatic**: lines **trophoblast** & covers **amniotic** cavity
  B. **Splanchnic**: covers **yolk**, then yolk becomes **secondary**
  - NB: The two are connected at caudal end of embryo by
    - **connecting stalk** (umbilical cord)

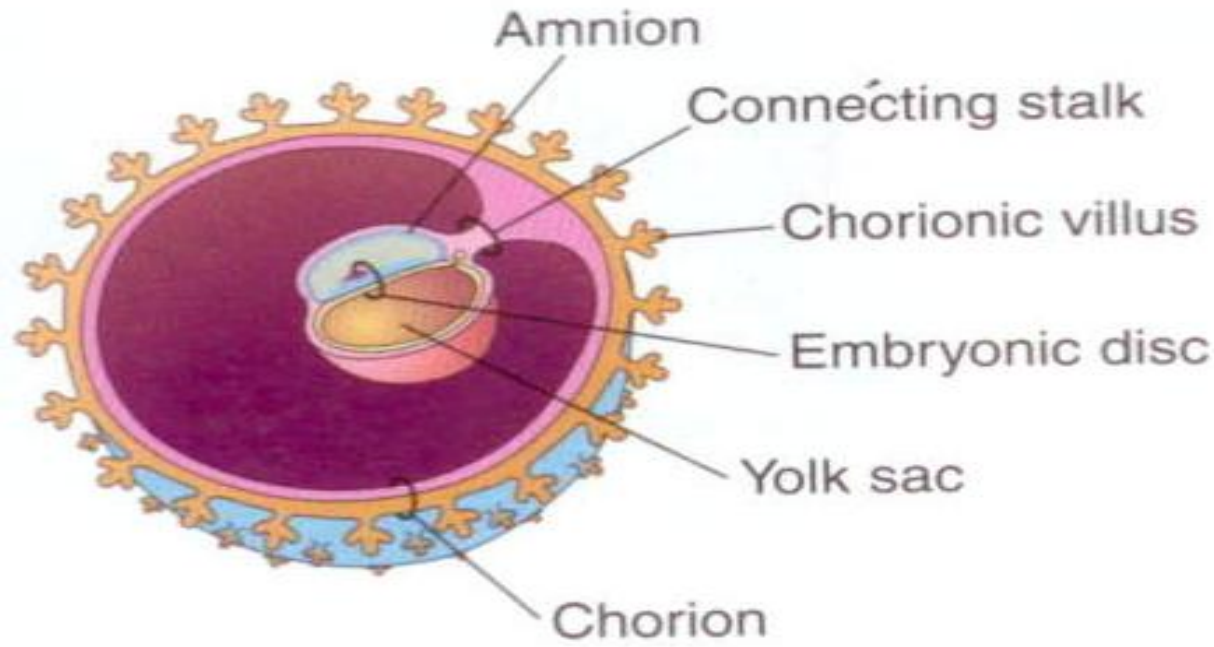


B



- **CHORION: The somatopleuric extra embryonic mesoderm with the trophoblast.**
  Forms the lining of chorionic cavity.

- **AMNION : The somatopleuric extra embryonic mesoderm with the amniotic cells .**
- **Forms the lining of amniotic cavity.**



REFERENCES

- **Langman's Medical Embryology: Chapter of Implantation + Second week of pregnancy**

ANY QUESTION





THANK ` U ’

"The important thing is not to stop questioning.

Curiosity has its own reason for existing."

